

Experiment 2: Develop a simple application to understand the concept of PAAS using Render and launch it

Aim

To develop and deploy a simple Flask web application using Render and understand how the Platform as a Service (PaaS) cloud model simplifies application deployment.

What is PaaS?

Platform as a Service (PaaS) is a cloud computing service that provides developers with a complete platform to build, test, deploy, and manage applications without worrying about the underlying infrastructure. The cloud provider manages servers, operating systems, networking, storage, and software updates, allowing developers to focus only on writing and deploying application code.

Benefits of PaaS

- No need to manage servers.
- Faster application deployment.
- Automatic software installation.
- Easy collaboration among developers.
- Scalable and cost-effective.

Examples of PaaS Providers

Some popular Platform as a Service providers are:

- Render
- Google App Engine
- Microsoft Azure App Service
- Heroku
- AWS Elastic Beanstalk
- Red Hat OpenShift

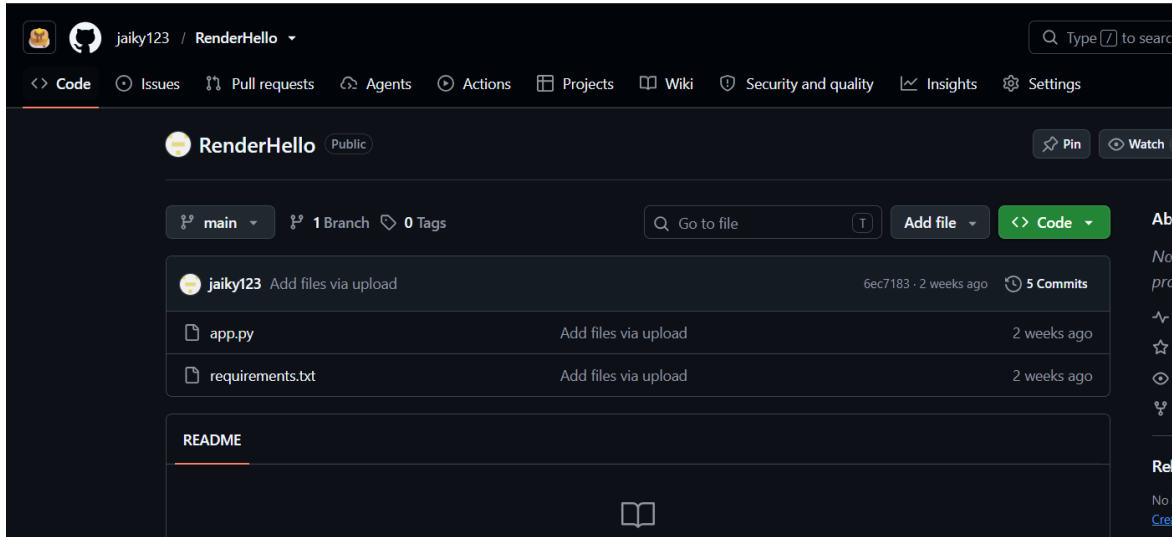
What is Render?

Render is a modern cloud Platform as a Service (PaaS) that allows developers to deploy web applications, APIs, static websites, and databases directly from Git repositories. It automatically installs the required software, creates the hosting environment, and provides a public URL for accessing the application.

Procedure

Step 1: Create a GitHub Repository

1. Sign in to GitHub.
2. Create a new public repository named **RenderHelloApp**.



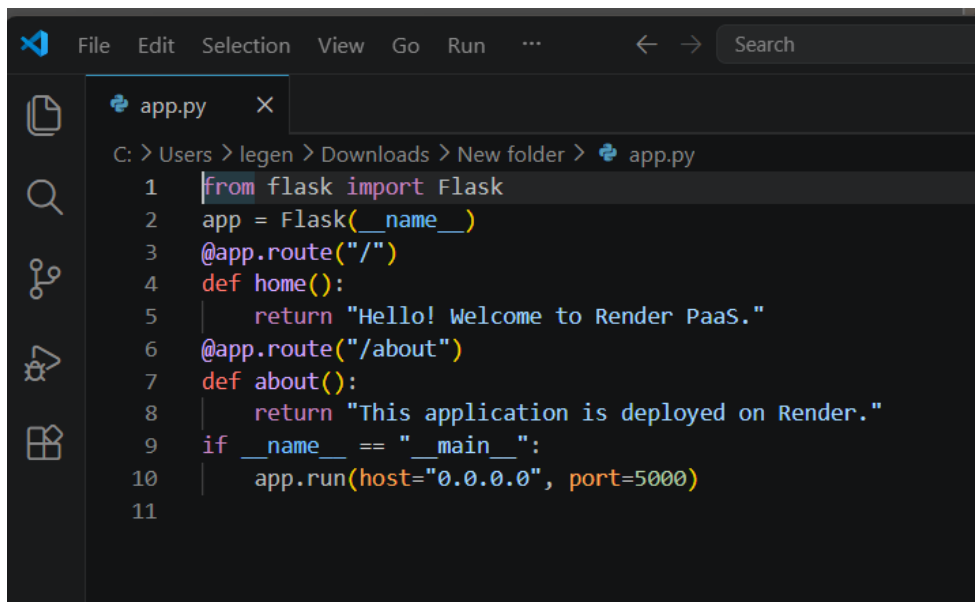
3. This repository will store the project files.

Step 2: Create the Project Files

1. Create a folder named **RenderHelloApp**.
2. Inside the folder, create:
 - app.py
 - requirements.txt

Step 3: Write the Flask Application

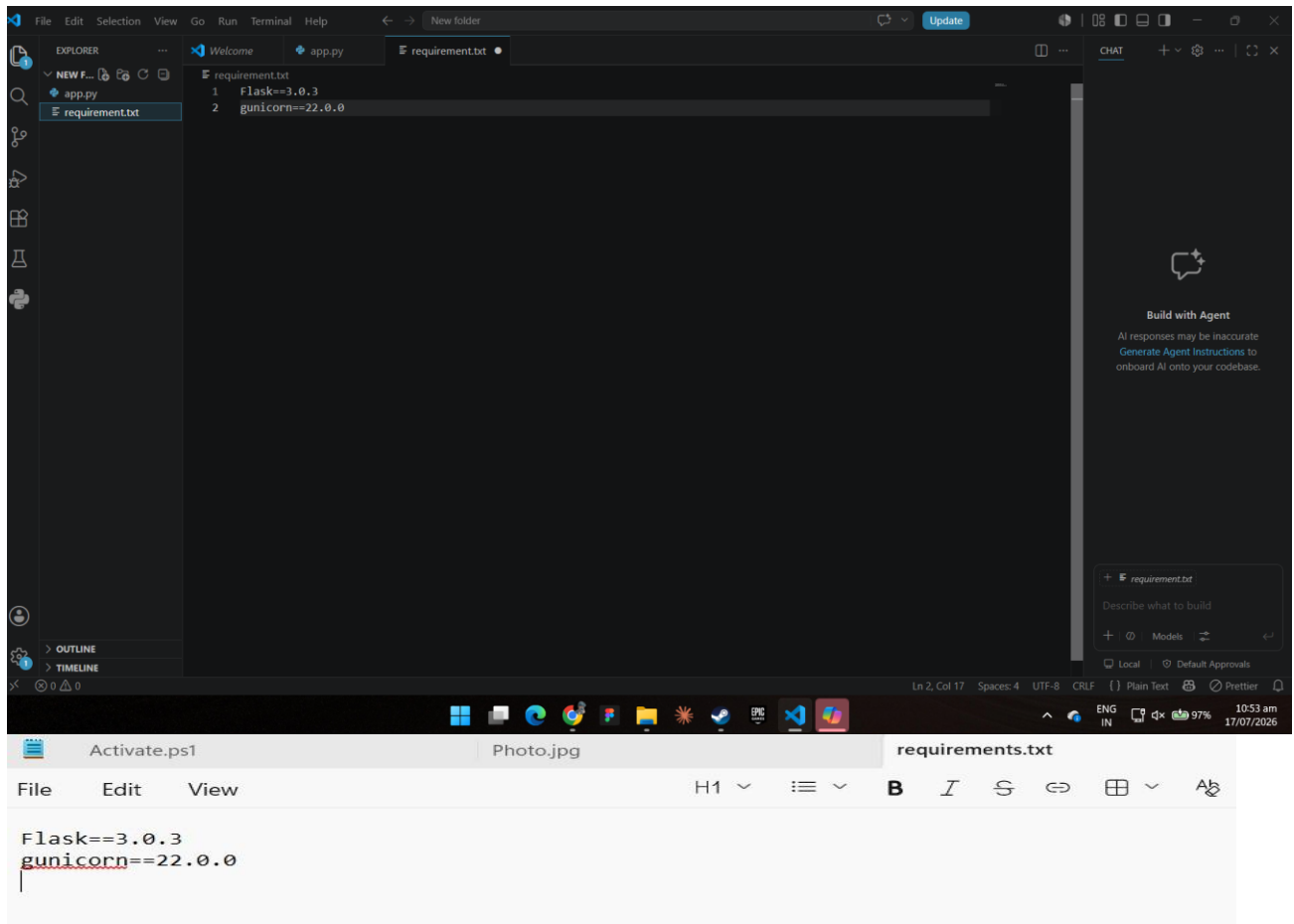
1. Open **app.py**.
2. Import the Flask library.
3. Create a Flask application object.
4. Define the Home (/) route.
5. Define the About (/about) route.
6. Run the application on port 5000.

A screenshot of a code editor window with a dark theme. The window title is 'app.py'. The file path in the breadcrumb is 'C: > Users > legen > Downloads > New folder > app.py'. The code is as follows:

```
1 from flask import Flask
2 app = Flask(__name__)
3 @app.route("/")
4 def home():
5     return "Hello! Welcome to Render PaaS."
6 @app.route("/about")
7 def about():
8     return "This application is deployed on Render."
9 if __name__ == "__main__":
10     app.run(host="0.0.0.0", port=5000)
11
```

Step 4: Add Project Dependencies

1. Open **requirements.txt**.
2. Add the required Python packages:
 - Flask
 - Gunicorn



Step 5: Upload the Project to GitHub

1. Upload **app.py** and **requirements.txt** to the repository.
2. Commit the changes.

Step 6: Create a Render Account

- Visit <https://render.com>.
- Sign in using your GitHub account.

Step 7: Deploy the Application

1. Select New Web Service.

The screenshot shows the Render dashboard for a project named 'RenderHello'. The left sidebar contains navigation options: Environment, RenderHello, Events, Settings, MONITOR (Logs, Metrics), and MANAGE (Environment, Shell, Scaling, Previews, Changelog, Invite a friend, Contact support, Render Status). The main content area displays the 'WEB SERVICE' details for 'RenderHello', including the Service ID, repository link, and a URL. A purple banner indicates that the free instance will spin down with inactivity. Below this, a deployment log shows a failed attempt on July 17, 2026, at 10:44 AM, with the error message 'Exited with status 1 while building your code.' The log also shows the cloning of the repository and the use of Python version 3.14.3. The bottom of the screen shows a Windows taskbar with various application icons and the system clock indicating 10:49 am on 17/07/2026.

dashboards.render.com/web/srv-d9crkf0js32c73e2tie0/deployments/dep-d9crkf0js32c73e2tjbg?r=2026-07-17%4005%3A13%3A38~2026-07-17%4005%3A15%3A55

My project / Production / RenderHello

WEB SERVICE
RenderHello Python3 Free Upgrade your instance →

Service ID: srv-d9crkf0js32c73e2tie0

jaiky123 / RenderHello main

https://renderhello-z1pa.onrender.com

ⓘ Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more. Upgrade now

July 17, 2026 at 10:44 AM Failed

6ed56cb Add files via upload Rollback

Exited with status 1 while building your code.
Read our docs for common ways to troubleshoot your deploy.

All logs Search logs Jul 17, 10:43 AM - 10:45 AM GMT+5:30

Jul 17

10:44:40 AM ==> Cloning from https://github.com/jaiky123/RenderHello

10:44:41 AM ==> Checking out commit 6ed56cbd0c3e43a4a72125f2219d592cc8396c33 in branch main

10:44:41 AM ==> Using Python version 3.14.3 (default)

10:44:41 AM ==> Docs on specifying a Python version: https://render.com/docs/python-version

10:49 am 17/07/2026

2. Connect the GitHub repository.

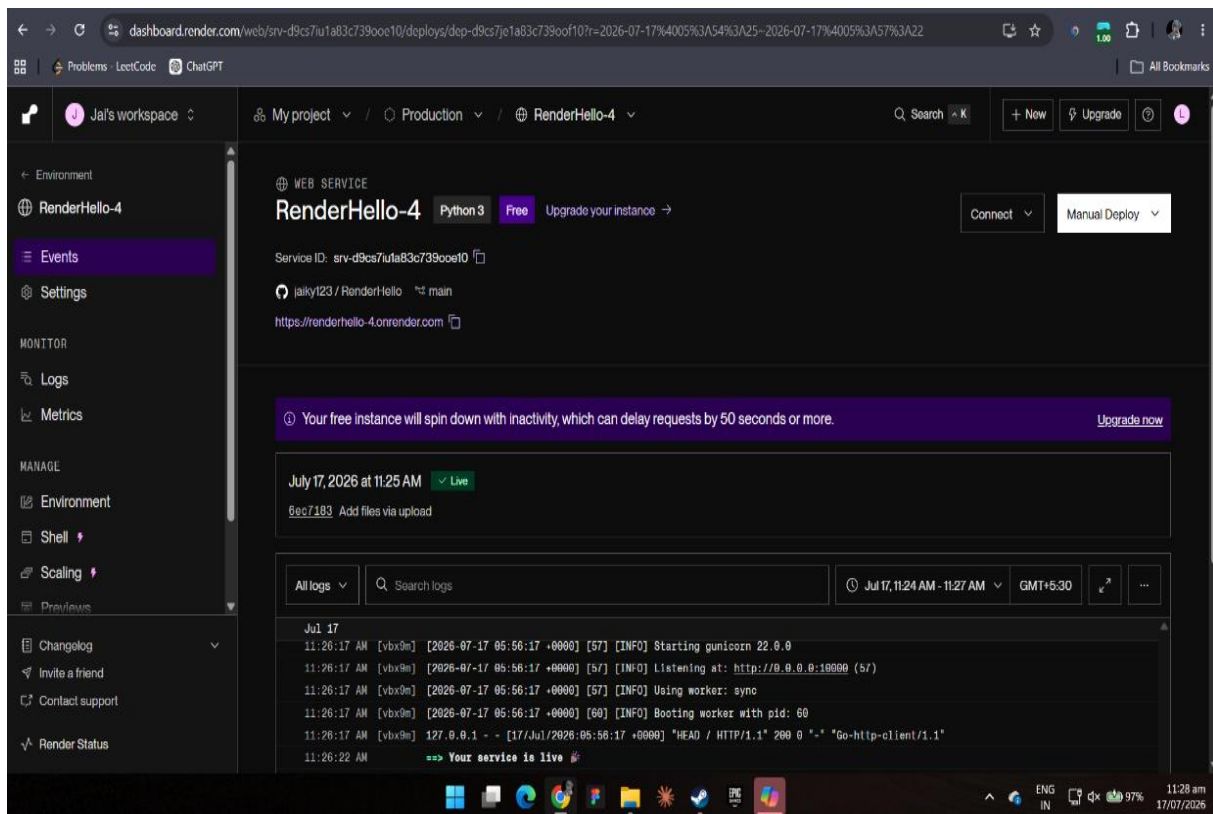
3. Configure the deployment:

- Environment: Python
- Build Command: pip install -r requirements.txt
- Start Command: gunicorn app:app

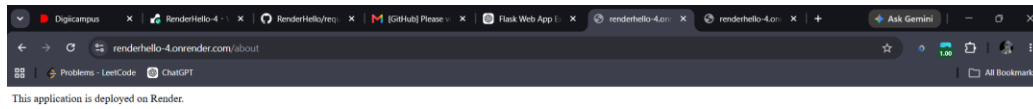
Click **Deploy Web Service**. Step

8: Verify the Deployment

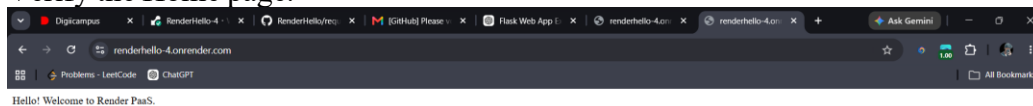
Wait until deployment is completed



1. the generated Render URL



2. Verify the Home page.



3. Open the /about page and verify the output

Conclusion

The experiment successfully demonstrated the deployment of a Flask web application using Render, a Platform as a Service (PaaS). The application was uploaded to GitHub, deployed on Render, and made accessible through a public URL without requiring any server configuration. This experiment helped in understanding how PaaS platforms simplify application hosting by managing the infrastructure while allowing developers to focus on application development.

